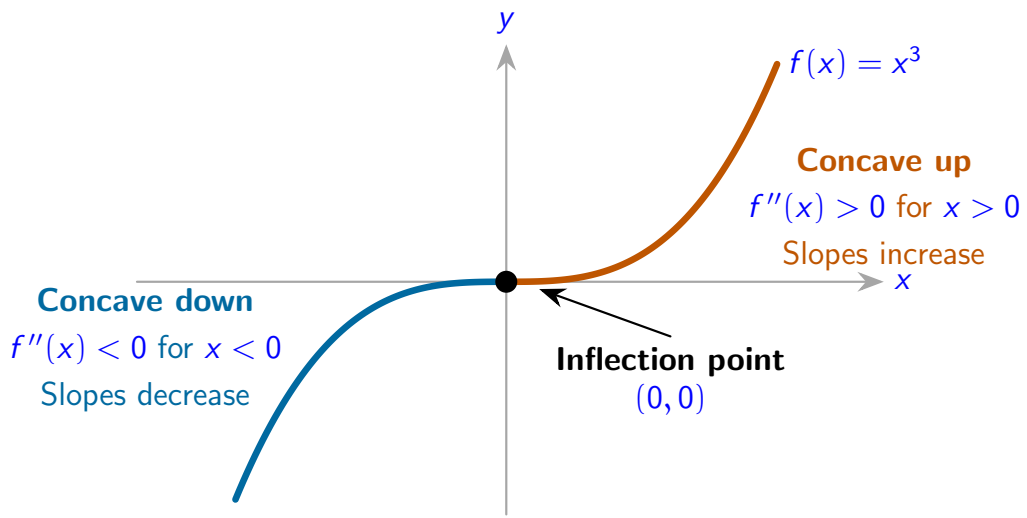
2627-MA140 Engineering Calculus

Week 06, Lecture 3
Maxima/Minima & Curve sketching

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A sketch of today's class...

1. Recall: Critical points
2. The First Derivative Test
   - Example
   - The Test
   - Example
3. Shape: Concave up and down
4. Inflection points
5. Second derivative test
6. Curve Sketching
7. Exercises

Slides on canvas, and niallmadden.ie/2627-MA140



Text-book references: Section 4.3 (Maxima and Minima), Section 4.5 (Derivatives and the Shape of a Graph) Link to book: https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)

Recall: Critical points

Definition: critical points

Let c in an point in the domain of a function f. We say that $x = c$ is a **critical point** of $f(x)$ if either

$$f'(c) = 0 \qquad \text{or} \qquad f'(c) \text{ does not exist.}$$

Important: If f has an **extremum** (that is, a local maximum or minimum) at $x = c$, then c must be a critical point of f.

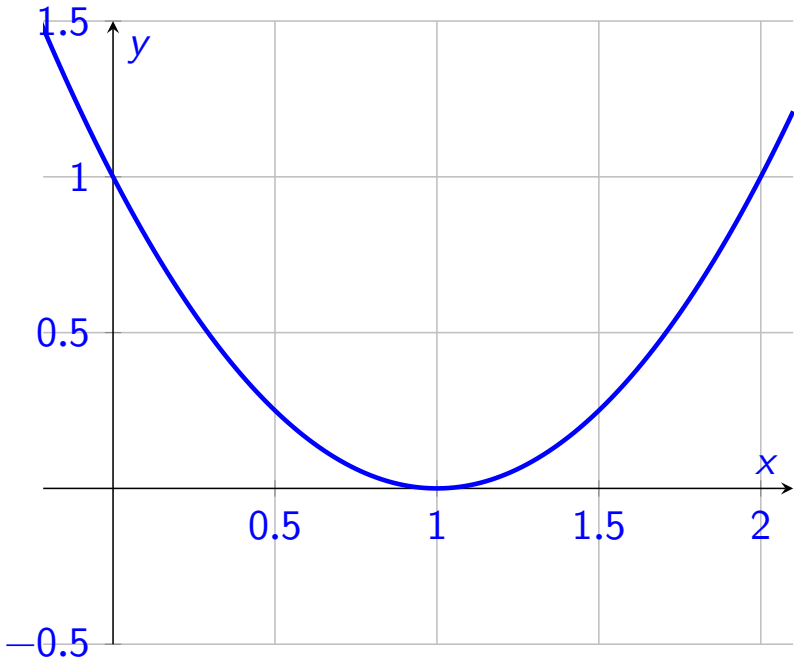
So, to find a maximum or minimum of f, it is enough to check at the critical points.

Note: All extrema are at critical points, but not all critical points correspond to a extrema.

Recall: Critical points

Example

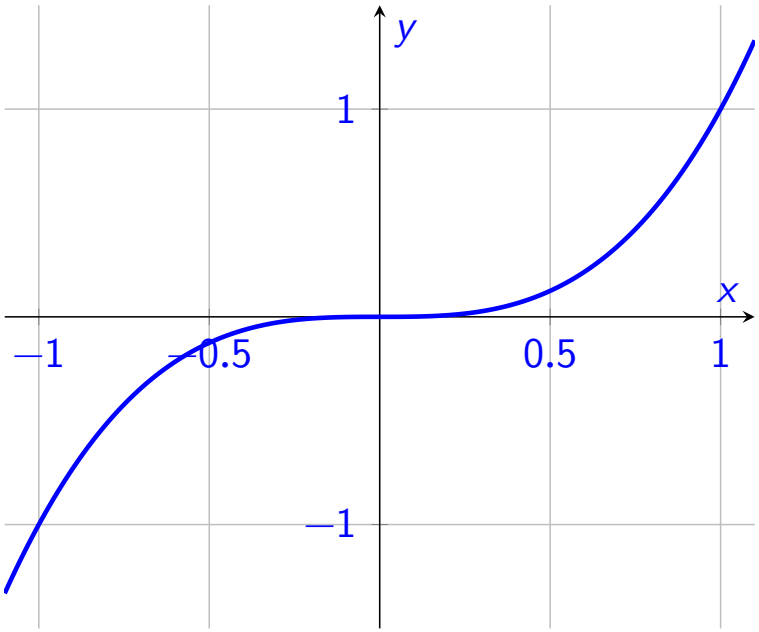
$f(x) = x^2 - 2x + 1$ has one critical point. Find it. Does it correspond to an extremum?



Recall: Critical points

Example

Find all critical points of $f(x) = x^3$. Do they correspond to extrema?

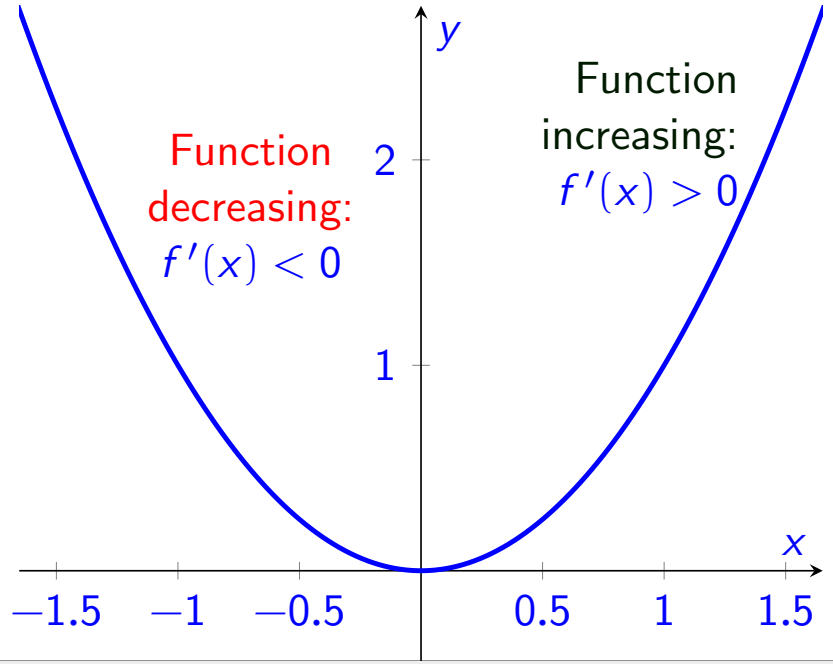


The First Derivative Test

Theorem (1st derivative test)

Suppose we have a function f, at that $f'(x)$ exists on an interval $\mathrm{I} = [a, b]$. Then

- If $f'(x) > 0$ for all $x \in \mathrm{I}$, then f is increasing on I.
- If $f'(x) < 0$ at all $x \in \mathrm{I}$, then f is decreasing on I.



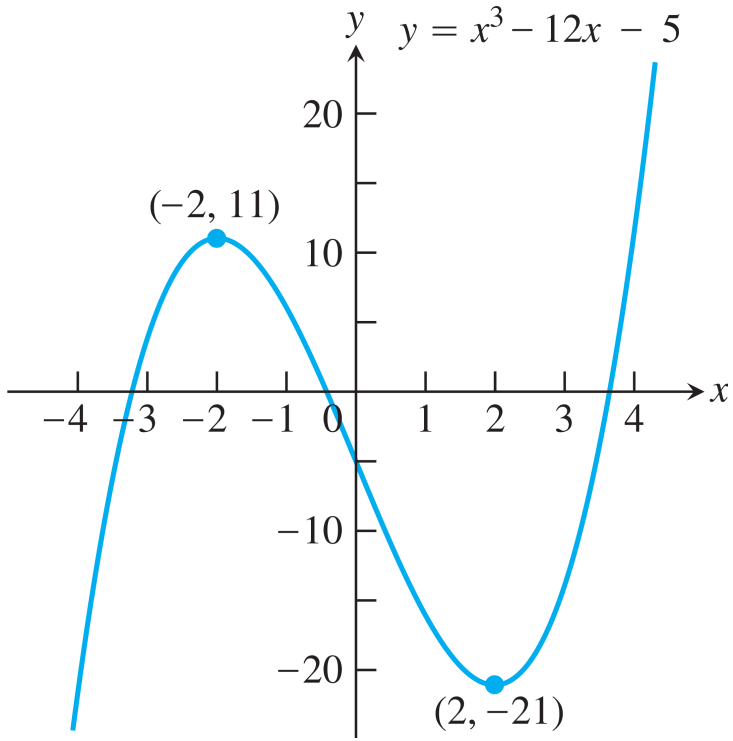
Example

Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the intervals on which f is increasing and decreasing

Idea: find $f'(x)$ and then solve for $f'(x) = 0$.

The critical points $c = -2$ and $c = 2$ of $f(x) = x^3 - 12x - 5$ subdivide the domain of f into intervals $(-\infty, -2), (-2, 2)$ and $(2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f at a convenient point in each subinterval.

	-2		2	
$3(x+2)$	$-$	$\bullet$	$+$	$+$
$x-2$	$-$		$-$	$\bullet$ $+$
$f'(x)$	$+$	$\bullet$	$-$	$\bullet$ $+$



Important:

- If $f(x)$ has a local minimum of $f(x)$ at $x = c$, then it switches from **decreasing** to **increasing**. That means, $f'(x)$ changes sign at $x = 2$. Therefore, $f'(c) = 0$.
- If $f(x)$ has a local maximum at $x = c$, we have that $f'(c) = 0$.

First Derivative Test for local maxima and minima

Suppose that c is a critical point of a differentiable function f.

1. If f' changes sign from positive when $x < c$ to negative when $x > c$, then $f(c)$ is a local maximum of f.
2. If f' changes sign from negative when $x < c$ to positive when $x > c$ then $f(c)$ is a local minimum of f.
3. If f' has the same sign for $x < c$ and $x > c$ then $f(c)$ is neither a local maximum nor a local minimum of f.

Example (Example 4.5.1 from textbook)

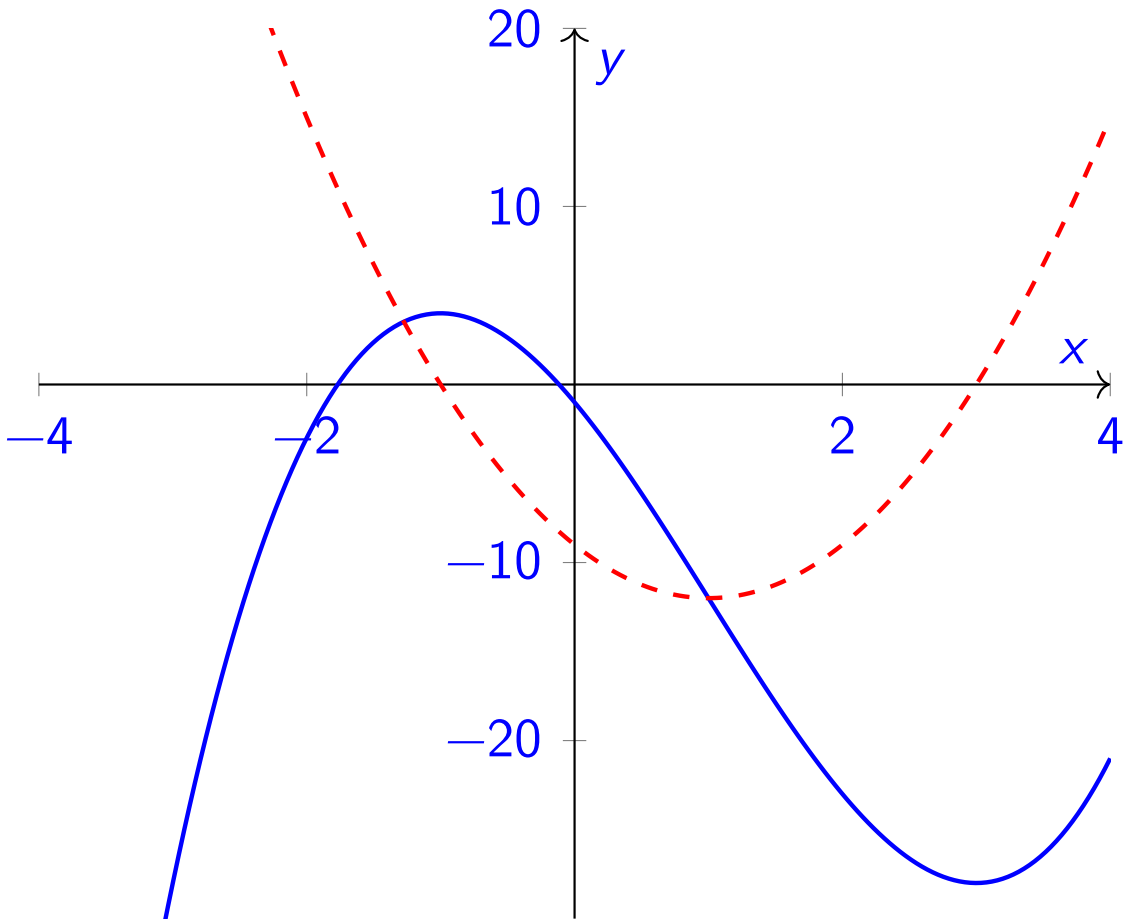
Use the first derivative test to find the location of all local extrema of $f(x) = x^3 - 3x^2 - 9x - 1$, and characterize them as maxima or minima.

1. Differentiate $f(x)$ to get $f'(x) = 3x^2 - 6x - 9$.
2. Solve for the critical points. Since $f'(x)$ is defined everywhere, we just need to solve $3x^2 - 6x - 9 = 0$. Simplifying, this is $x^2 - 2x - 3 = 0$. That factorizes as $f'(x) = (x+1)(x-3)$, which has two zeros: at $x = -1$, and $x = 3$.
3. Now we need to know how f' is changing sign at these points. Check the text-book for a technical approach, we'll use a simple one.

4. By calculation (e.g., with a calculator), we'll check $x = -1$. We see $f'(-1.1) = 1.23$ and $f'(-0.9) = -1.17$. So f' changes from **positive** to **negative** at $x = -1$, so we have a **local maximum**.

5. Similarly, we'll check $x = 3$. We see $f'(2.9) = -1.17$ and $f'(3.1) = 1.23$. So f' changes from **negative** to **positive** at $x = 3$, so we have a **local minimum** there.

A plot of $f(x)$ and $f'(x)$ (but which is which??)



Shape: Concave up and down

We now introduce some terminology to help us describe the shape of a curve of a function.

Definition

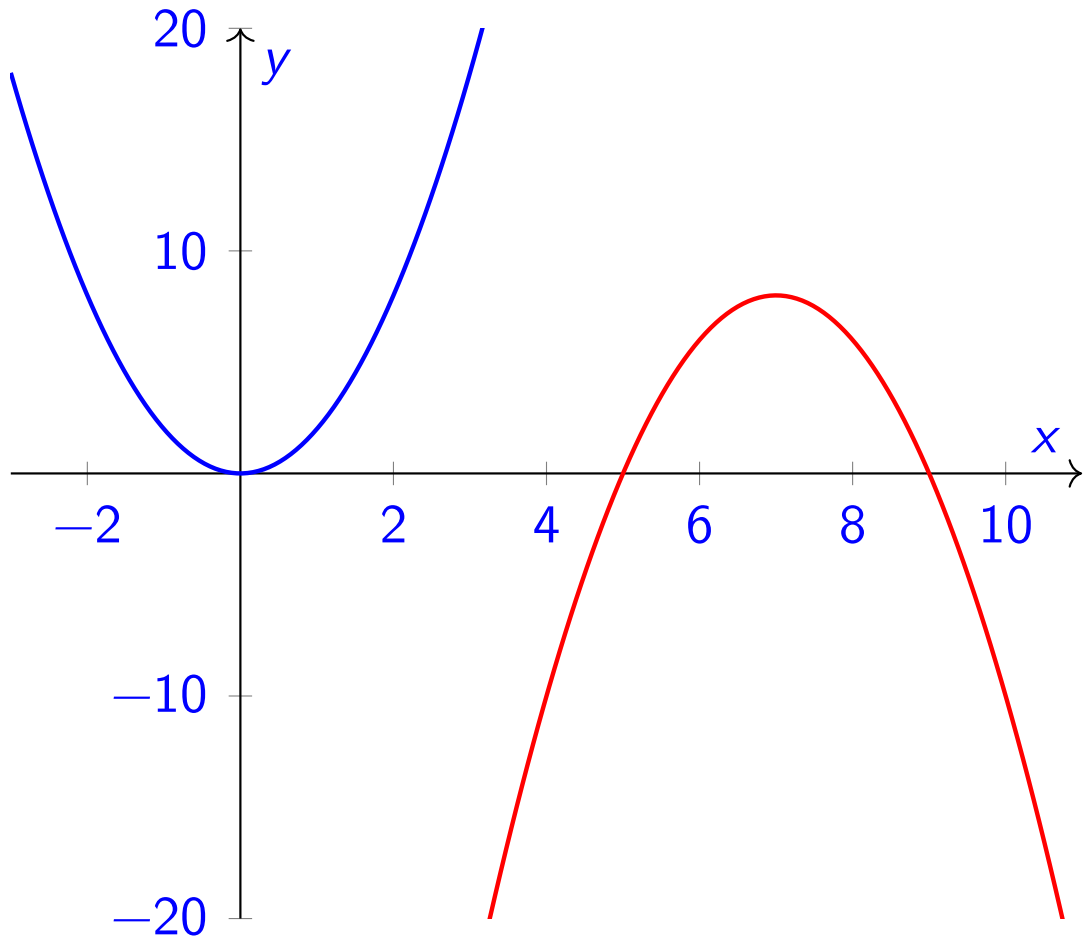
The graph of a differentiable function $y = f(x)$ is:

- **concave up** on an open interval (a, b) if f' is increasing on (a, b);
- **concave down** on an open interval (a, b) if f' is decreasing on (a, b)

Note:

- If the graph of f is **concave up** ("cup"), it is **above** its tangents.
- If the graph of f is **concave down**, it is **below** its tangents.

Shape: Concave up and down



Shape: Concave up and down

Relating concavity to f''

If f is a function such that $f''(x)$ exists on the interval (a, b). If for all $x \in (a, b)$ ...

- If $f''(x) > 0$, the graph of f is **concave up**
- If $f'' < 0$, the graph of f is **concave down**

Example: $f(x) = x^2$ is concave up (for all x) and $g(x) = -x^2$ is concave down.

Inflection points

Definition: infection point

A **point of inflection** is a point at which the concavity of a function changes.
At such a point, either $f''(x)$ is zero or does not exist.

Example

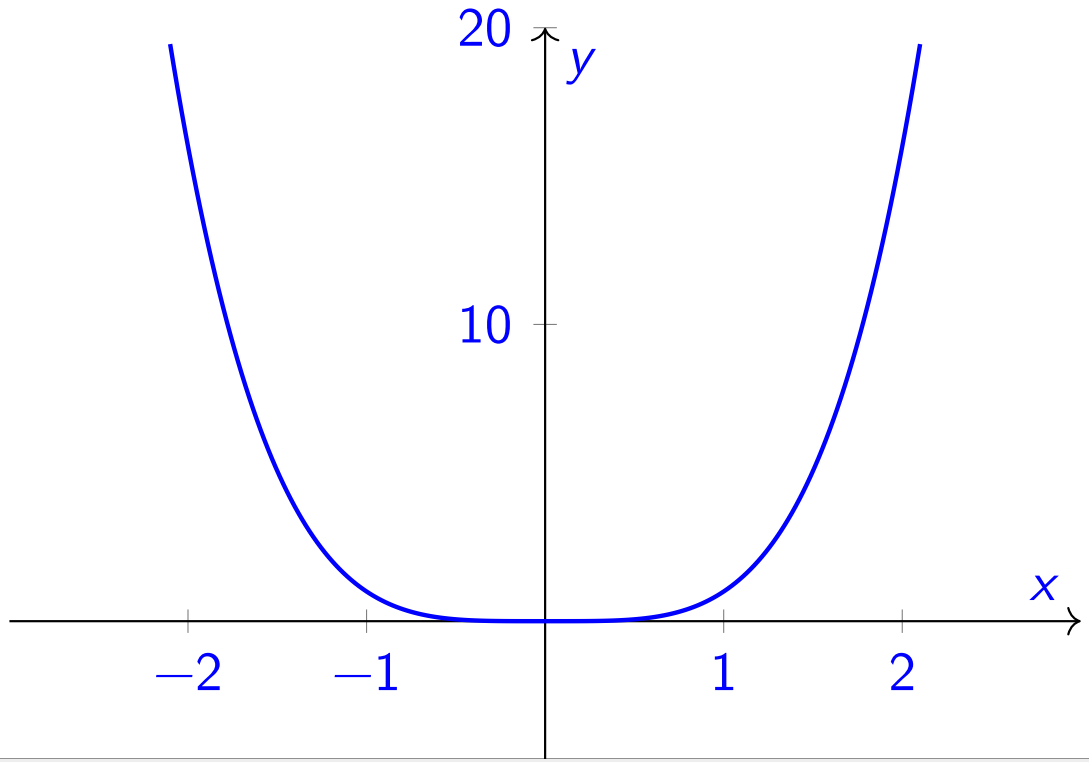
Find a point of inflection of the graph of $f(x) = x^3$.

Inflection points

Warning: Having $f''(x) = 0$ does not necessarily mean that f has an inflection point at x.

Example

The curve $y = x^4$ has no inflection point at $x = 0$. Even though $y'' = 12x^2$ is zero there, it does not change sign.

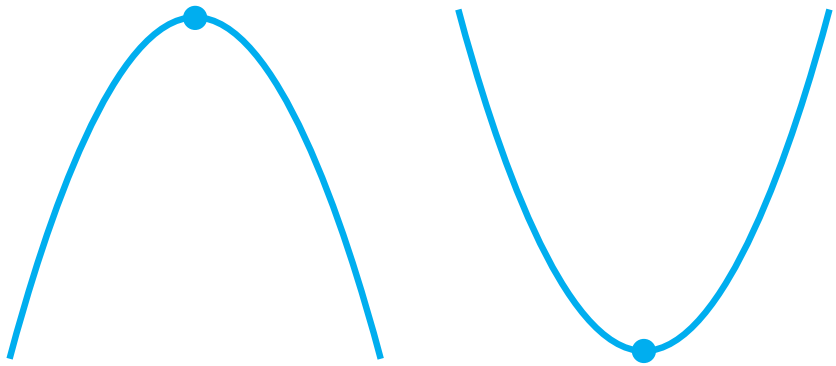


Second derivative test

Second Derivative Test

Suppose that f'' is continuous on an interval that contains c.

- If $f'(c) = 0$ and $f''(c) < 0$, then f has a **local max** at $x = c$.
- If $f'(c) = 0$ and $f''(c) > 0$, then f has a **local min** at $x = c$.
- If $f'(c) = 0$ and $f''(c) = 0$, then the test is inconclusive. The function f may have a local max, a local min, or neither.



$f' = 0, f'' < 0$
$\Rightarrow$ local max

$f' = 0, f'' > 0$
$\Rightarrow$ local min

Second derivative test

Example

Find and classify the critical and infection points of

$$f(x) = 4x^3 - 21x^2 + 18x + 6.$$

- We have $f'(x) = 12x^2 - 42x + 18$.

  When $f'(x) = 0$, we have

  $$12x^2 - 42x + 18 = 0 \Leftrightarrow 2x^2 - 7x + 3 = 0$$
  $$\Leftrightarrow (2x - 1)(x - 3) = 0.$$

  So the critical points are at $x = 1/2$ and $x = 3$.
- Next $f''(x) = 24x - 42$ so

  $$f''(\frac{1}{2}) = 24(\frac{1}{2}) - 42 = 12 - 42 = -30 < 0,$$

  which means there is a local **maximum** at $x = 1/2$.

Second derivative test

Also,

$$f''(3) = 24(3) - 42 = 72 - 42 = 30 > 0.$$

so it has a local **minimum** at $x = 3$.

. . . . . .

Now, recall that $f''(x) = 24x - 42$. Thus,

$$f''(x) = 0 \Leftrightarrow x = \frac{42}{24} = \frac{7}{4}.$$

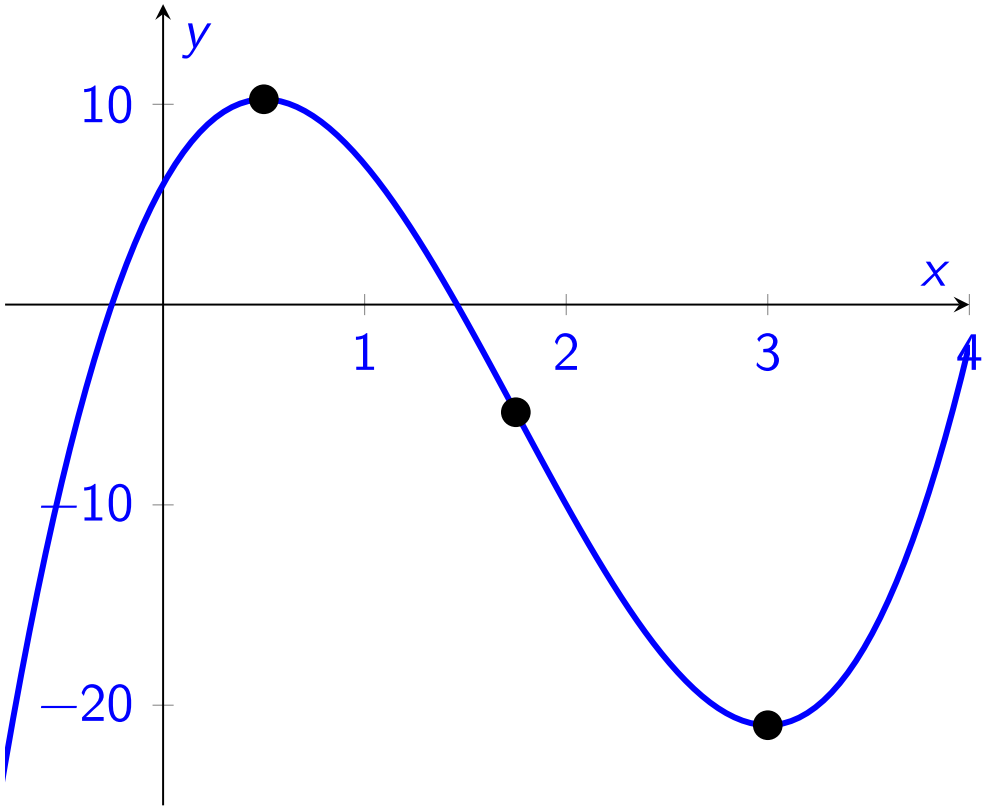
Note that

$$x < \frac{7}{4} \implies f''(x) < 0$$

$$x > \frac{7}{4} \implies f''(x) > 0.$$

Therefore, $f(x)$ has a point of inflection at $x = \frac{7}{4}$.

Second derivative test



Curve Sketching

In order to roughly **sketch the graph** of a function, f, we can use the following steps:

1. Compute $f'(x)$ and find the **critical points** and inflection points of f. Find the corresponding y-value of these points.
2. Compute $f''(x)$, and use the second derivative test.
3. Make a table showing the intervals on which f is increasing and/or decreasing, and where f is concave up and/or concave down.
4. Plot some specific points (e.g. local max/min, points of inflection, intercepts) and sketch the general shape of the graph of f.

Curve Sketching

Example

Sketch the graph of the function $f(x) = x^4 - 4x^3 + 10$

Curve Sketching

Step 3: Make table to find intervals on which f is increasing/decreasing and on which f is concave up and concave down

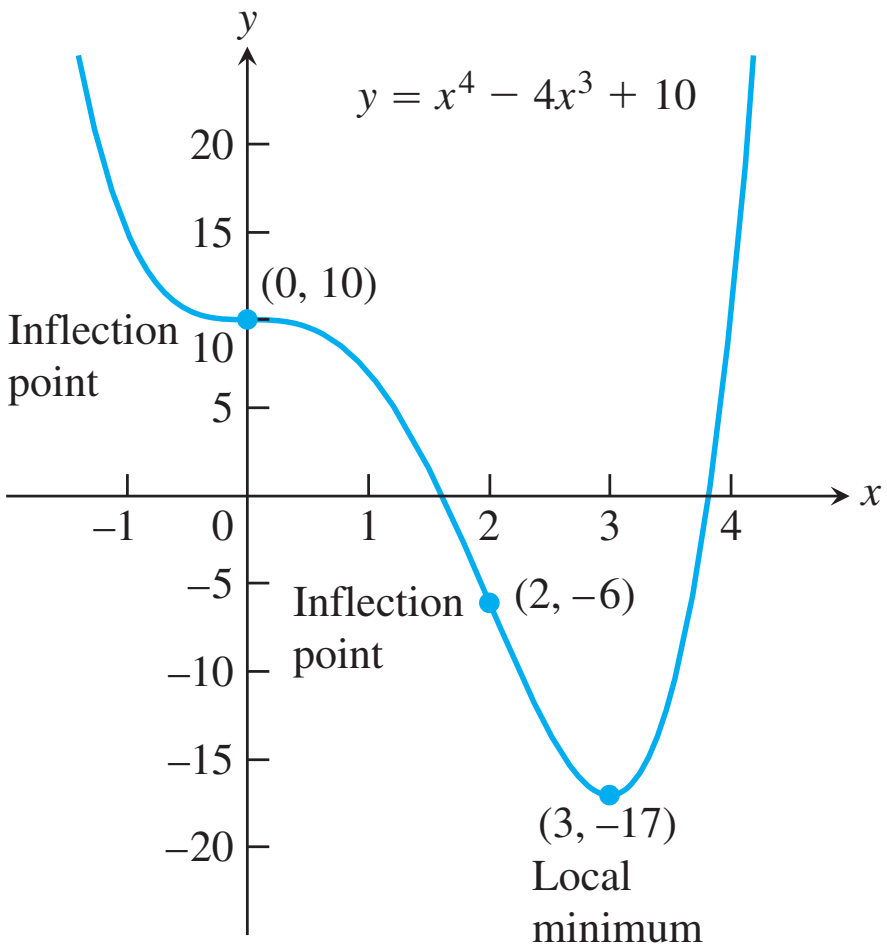
		0		2		3	
$4x^2$	$+$	$\bullet$	$+$		$+$		$+$
$x-3$	$-$		$-$		$-$	$\bullet$	$+$
$f'(x)$	$-$	$\bullet$	$-$		$-$	$\bullet$	$+$
$12x$	$-$	$\bullet$	$+$		$+$		$+$
$x-2$	$-$		$-$	$\bullet$	$+$		$+$
$f''(x)$	$+$	$\bullet$	$-$	$\bullet$	$+$		$+$

Curve Sketching

Step 4: Plot specific points (such as local max/min, points of inflection, intercepts) - and sketch general shape of graph of f

Curve Sketching

Step 5: Plot specific points (such as local max/min, points of inflection, intercepts) - and sketch general shape of graph of f



Exercises

Exercise 6.3.1 : From 23/24 Exam

Let $f(x) = \ln(x^2 + 1)$.

(i) Find all critical point(s) of f and determine whether f has a local minimum, local maximum or neither.

(ii) Determine the interval on which f is increasing.

(iii) Determine the interval on which f is decreasing.

Exercises

Exer 6.2.2 (Based on 2024/2025 Exam)

Let $f(x) = x^2 e^x$.

1. Find all critical points of f, and determine if f has a local minimum, maximum or neither.
2. Determine the two interval on which f is increasing, and the one where f is decreasing.
3. Find all point(s) of inflection for the graph of $f(x)$.
4. Give a rough sketch the graph of $f(x)$ (your axes need not necessarily have the same scale).